

Project Report Of DESIGN THINKING

(HSMC 102)

**Bridging the Skill Gap: A Smart Analyzer for Career
Readiness**

BTECH CSE-II SEM

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School of Engineering ,Design & Automation

GNA UNIVERSITY

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Date: 01-05-2026

CERTIFICATE

This is to certify that the project report titled “**Bridging the Skill Gap: A Smart Analyzer for Career Readiness**” is a Bonafide work carried out by **Zeenat Ansari (GU-2025-3632) and Avni Mandal (GU-2025-3641)** of **B.Tech – CSE, 2nd Semester**, under my supervision and guidance, in partial fulfilment of the requirements of the subject Design Thinking during the academic year 2024-2025. This project has not been submitted earlier, either partly or wholly, for the award of any degree or diploma.

Signature of Guide Name:

Er. Gagandeep Singh Sharma

Designation: Assistant Professor, GNA University

Date: 06-05-2025

Place: GNA University

DECLARATION

I hereby declare that the project titled “**Bridging the Skill Gap: A Smart Analyzer for Career Readiness**” submitted to the Department of Computer Science & Engineering is a record of original work done by me under the guidance of **Er. Gagandeep Sharma** and it has not been submitted elsewhere for any other purpose.

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Abstract

The Skill Gap Analyzer project focuses on solving a common problem faced by students today — they are learning, but not the skills that industries actually require. Many students are unaware of in-demand skills, depend mainly on theoretical knowledge, and lack proper career guidance. Because of this, they often feel confused, lose confidence after repeated failures, and struggle to find suitable job opportunities.

This project uses **the Design Thinking** approach to understand user problems and develop a practical solution. Through research and observations, key issues such as lack of awareness, theory vs practical gap, and absence of clear learning paths were identified. Based on these insights, a system was designed to help users analyze their **skills and identify** what they need to improve.

The system works by taking input from users in the form of skills or resumes. It then analyzes these skills, compares them with industry requirements, and identifies the missing skills. After that, it provides personalized learning roadmaps and suggestions. To make the process more engaging, features like adaptive tests, progress tracking, and gamification (leaderboards and challenges) are included.

Overall, the project provides a clear direction to students by helping them understand their skill gaps and guiding them toward improvement. It highlights the importance of practical learning and shows how technology can be used to improve employability.

Introduction

Background:

In recent years, education has become more accessible due to the growth of online learning platforms and digital resources. Students now have access to a wide range of courses and materials that allow them to learn anytime and anywhere. However, despite this availability, a major problem still exists — students often do not know which skills are actually required by industries. Most learners focus on theoretical knowledge provided by colleges, while practical and job-oriented skills are often ignored.

This creates a gap between what students learn and what companies expect. Many students feel confused about their career path and are unable to identify which skills they should develop. Lack of proper guidance, awareness, and structured learning paths makes it difficult for them to prepare for real-world challenges.

Relevance:

The problem of the skill gap is becoming increasingly important in today's competitive world. Students spend a lot of time learning skills that are not in demand, which leads to frustration and loss of confidence. At the same time, companies struggle to find candidates who are job-ready and possess the required skills.

This mismatch not only affects individual careers but also impacts the overall economy by increasing unemployment and underemployment. Therefore, it is important to develop a system that can guide students in the right direction and help them focus on relevant skill development.

Objectives:

The main objective of this project is to identify the gap between the skills users currently have and the skills required in the industry. It also aims to provide proper guidance and personalized suggestions to help users improve their skills. Another objective is to create a simple and user-friendly system that makes skill analysis easy and effective. The project also focuses on making learning more engaging through structured roadmaps and interactive features.

Scope and Limitations:

This project mainly focuses on students and beginners who need guidance in skill development. It is designed as a web-based system that analyzes user skills and provides recommendations based on predefined data. The system can be applied to different domains and can be further expanded in the future.

However, the current system has some limitations. It relies on basic skill comparison and does not include advanced AI-based analysis. It also does not use real-time industry data, which may affect the accuracy of recommendations. Despite these limitations, it provides a strong foundation for improving skill awareness and career readiness.

Design Thinking Process

3.1 Empathize

Stakeholder/User Research:

At the beginning of the design thinking process, we focused on understanding the real problems faced by students and job seekers. We conducted informal surveys and discussions to identify the primary users of our system. These included college students, recent graduates, and beginners who are trying to build their careers but are unsure about the skills required in their chosen fields.

During this research, we found that most students are confused about which skills are actually needed in industries. They often rely on theoretical knowledge from college but lack practical exposure. Many of them also mentioned that they do not have proper guidance or a clear roadmap for skill development. Some students even feel demotivated after facing repeated rejections due to missing skills.

We also considered the perspective of companies and industry expectations. It was observed that companies are looking for job-ready candidates with practical skills, but students are not fully prepared to meet these expectations.

Interviews/Observations

To gain deeper insights, we interacted with a few students and observed their learning patterns and challenges.

- **Rahul** – A final-year student who has good academic knowledge but is unsure about which technical skills to focus on for placements.
- **Priya** – A beginner learner who takes online courses but feels confused due to lack of direction and structured guidance.
- **Aman** – A job seeker who has faced multiple rejections because his skills do not match industry requirements.

From these interactions, we observed that students spend time learning random or less relevant skills, which leads to wasted effort and low confidence. They also struggle to track their progress and do not know how to improve effectively. These insights helped us clearly understand the need for a system that can guide users in identifying and improving their skills.

3.2 Define

Problem Statement

“How might we help students and job seekers identify their skill gaps and guide them with a clear and structured path to achieve industry-relevant skills?”

Common Challenges Faced:

During our research and user interactions, we identified several common challenges faced by students and job seekers while developing their skills. These challenges highlight the need for a system like the Skill Gap Analyzer.

One of the major issues observed was the lack of awareness. Many students are not aware of which skills are actually required in the industry. They often follow outdated or irrelevant learning paths without understanding current market demands. This leads to confusion and inefficient use of time and effort.

Another important challenge is the gap between theoretical knowledge and practical skills. Students mainly focus on academic learning, which does not always match real-world requirements. As a result, even after completing courses, they struggle to apply their knowledge in practical situations.

We also found that there is a lack of proper career guidance. Students do not have a clear roadmap that tells them what to learn next or how to progress step by step. Without direction, they often switch between different skills without mastering any of them.

A common problem faced by many users is low confidence and motivation. Repeated job rejections or failure to meet skill requirements can make students feel discouraged. This affects their willingness to continue learning and improving.

Another challenge is the absence of personalized learning paths. Most online platforms provide general courses, but they do not analyze individual skill levels or suggest what specifically needs improvement. This makes learning less effective and less targeted.

Finally, we observed that students find it difficult to track their progress and measure improvement. Without proper feedback or evaluation, they are unable to understand whether they are moving in the right direction.

Major Needs of Users:

Based on our research and understanding of the challenges faced by students, we identified several important needs that must be addressed to make skill development more effective and meaningful.

One of the primary needs is clear skill awareness and guidance. Users need to understand which skills are required in their desired career paths and how their current skills compare to industry expectations. Without this clarity, students often feel lost and unsure about what to learn next.

Another important need is personalized learning recommendations. Every user has a different skill level and background, so a one-size-fits-all approach does not work. Users require customized suggestions and step-by-step roadmaps that guide them from their current level to their desired goal.

Users also need a simple and user-friendly interface. The system should be easy to use, with clear navigation and minimal confusion. A well-designed interface helps users focus on learning rather than struggling with the system itself.

In addition, there is a strong need for structured and flexible learning paths. Users should be able to learn at their own pace while following a clear progression from basic to advanced skills. This helps in building confidence and ensures continuous improvement.

Another key requirement is progress tracking and feedback. Users need to see how much they have improved over time. Regular feedback helps them stay motivated and understand whether they are moving in the right direction.

Finally, users benefit from engaging and interactive learning experiences. Features like quizzes, challenges, and leaderboards can make the process more interesting and motivate users to stay consistent in their learning journey.

3.3 Ideate

Understanding the needs and challenges of users plays a very important role in generating effective solutions. In this stage, the goal was to explore different ideas that could help solve the problem of the skill gap.

Various solutions were considered, such as online learning platforms, resume analysis tools, and career guidance systems. However, most existing solutions either provide information without direction or lack personalization.

After evaluating different ideas, we selected the concept of a Skill Gap Analyzer. This system not only identifies missing skills but also provides a clear and structured roadmap for improvement. The idea also includes interactive and gamified features to make learning more engaging and effective.

Brainstorming:

During the ideation stage, we explored different ideas that could help solve the problem of the skill gap among students. The main focus was to think of solutions that not only identify missing skills but also guide users effectively in improving them.

We considered features such as a system that can analyze resumes and automatically extract user skills, along with a tool that compares these skills with industry requirements. Another idea was to include personalized learning paths that guide users step-by-step from beginner to advanced levels. We also explored adding adaptive tests that adjust according to the user's skill level, helping in more accurate evaluation.

To make the platform more engaging, we thought of incorporating gamification elements like quizzes, challenges, leaderboards, and reward systems. This would motivate users to stay consistent and make the learning process more interactive. Additionally, we discussed the importance of progress tracking so that users can monitor their improvement over time and stay motivated.

Mind Maps/Sketches:

To organize our ideas, we created simple mind maps and rough sketches that grouped features based on their purpose and usability. These included sections like skill analysis, gap detection, learning roadmap, and user engagement.

The sketches helped us visualize how the system would work step-by-step, starting from user input to final recommendations. It also helped in understanding the user flow and making the design more structured and easy to use.

Selected Idea Rationale:

After analyzing all the ideas, we selected the concept of a Skill Gap Analyzer with personalized guidance and gamification. This idea was chosen because it provides both analysis and direction, which is missing in most existing solutions.

The selected features were those that could create the highest impact while still being practical to implement. Instead of giving general suggestions, the system focuses on personalized recommendations and structured learning paths. This approach ensures that users not only understand their skill gaps but also get clear guidance on how to overcome them.

3.4 Prototype

Description:

We developed a web-based prototype of the Skill Gap Analyzer that focuses on helping users easily understand their current skills and identify what they need to improve. The main aim was to keep the system simple, interactive, and useful for students and job seekers.

The prototype allows users to enter their skills or upload their resume. Based on this input, the system analyzes the skills and compares them with industry requirements. It then identifies the missing skills and displays them clearly. Along with this, the system provides personalized learning roadmaps that guide users step-by-step on how to improve their skills.

To make the platform more engaging, features like adaptive quizzes and gamification elements such as leaderboards and challenges are included. These features help users stay motivated and actively participate in their learning process. The system also includes progress tracking so users can monitor their improvement over time.

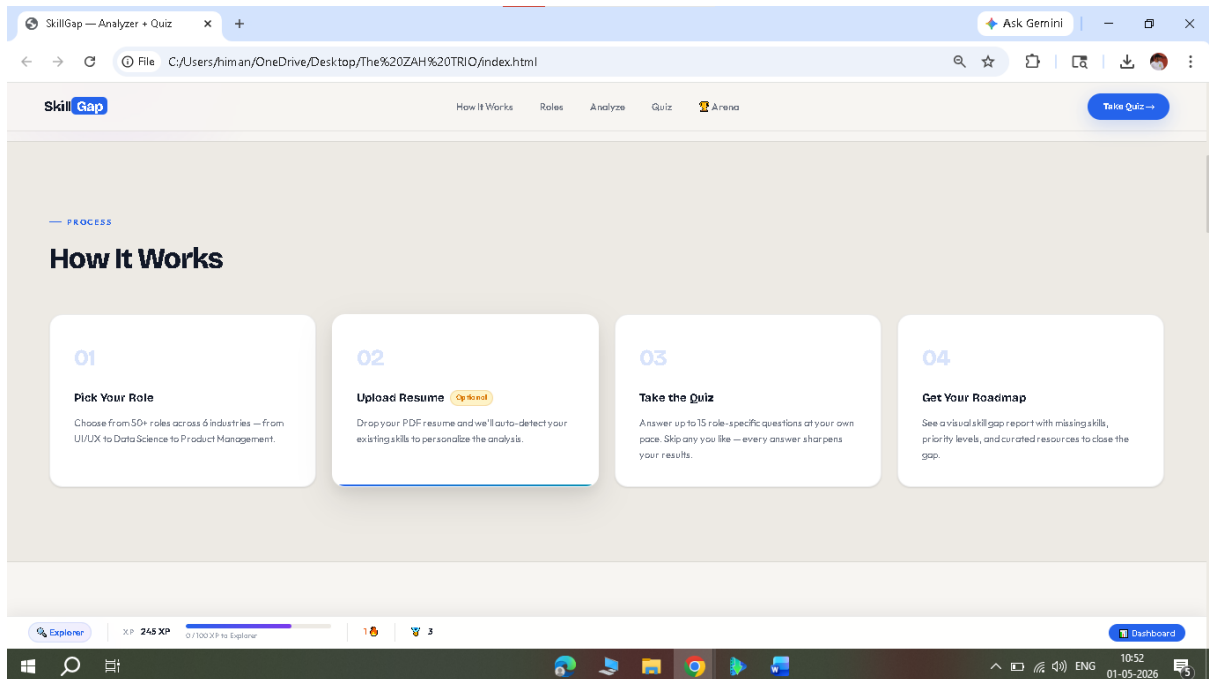
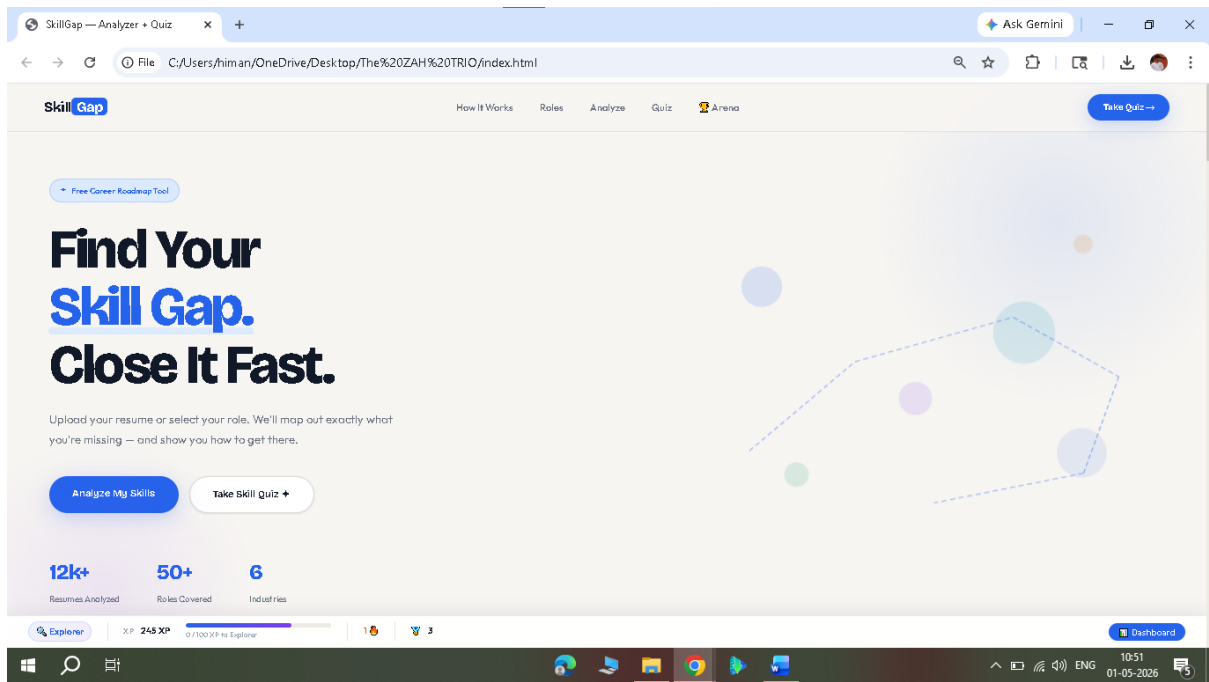
Overall, the prototype is designed to provide a clear and structured path from identifying skill gaps to achieving career goals.

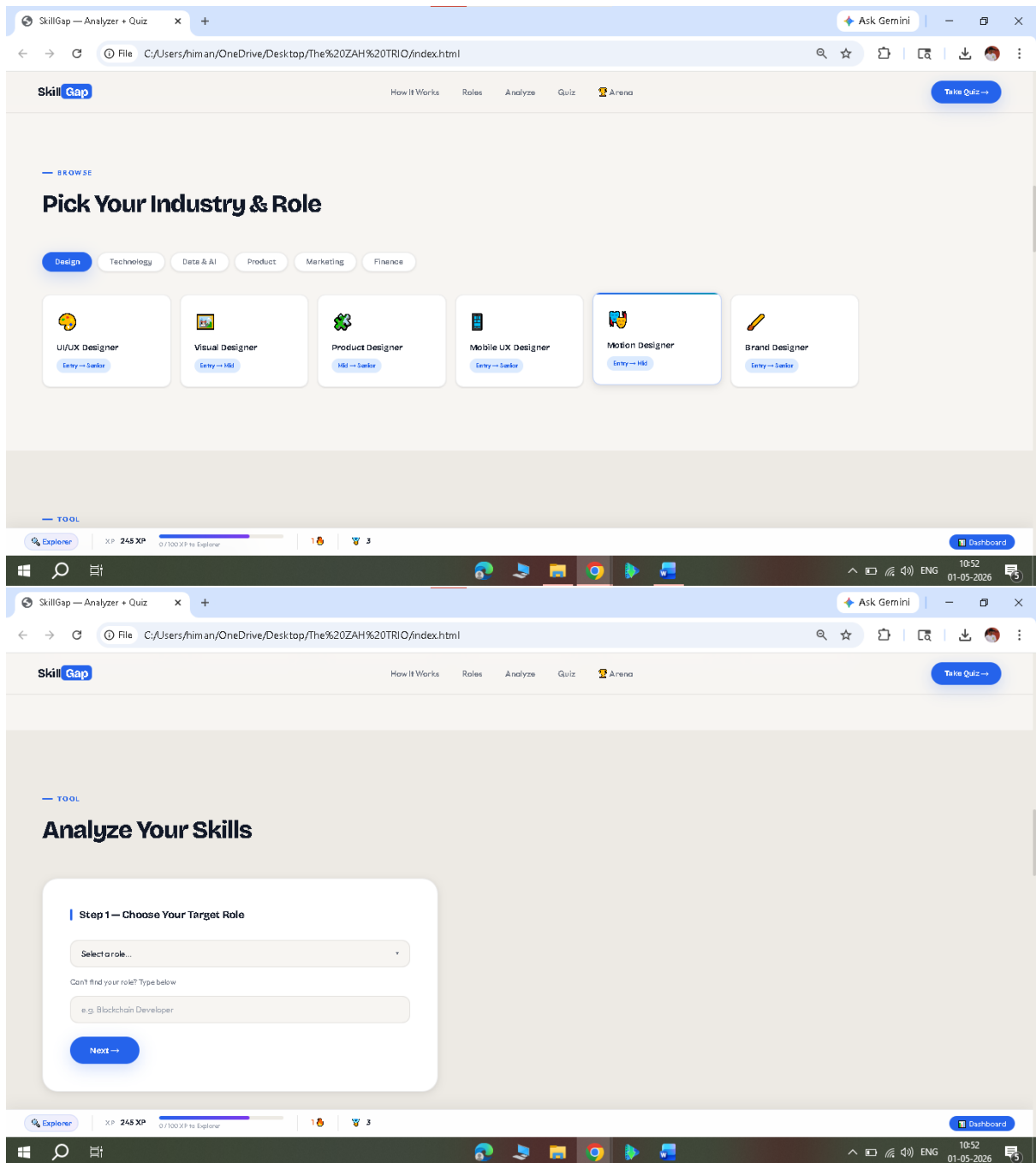
Tools and Materials:

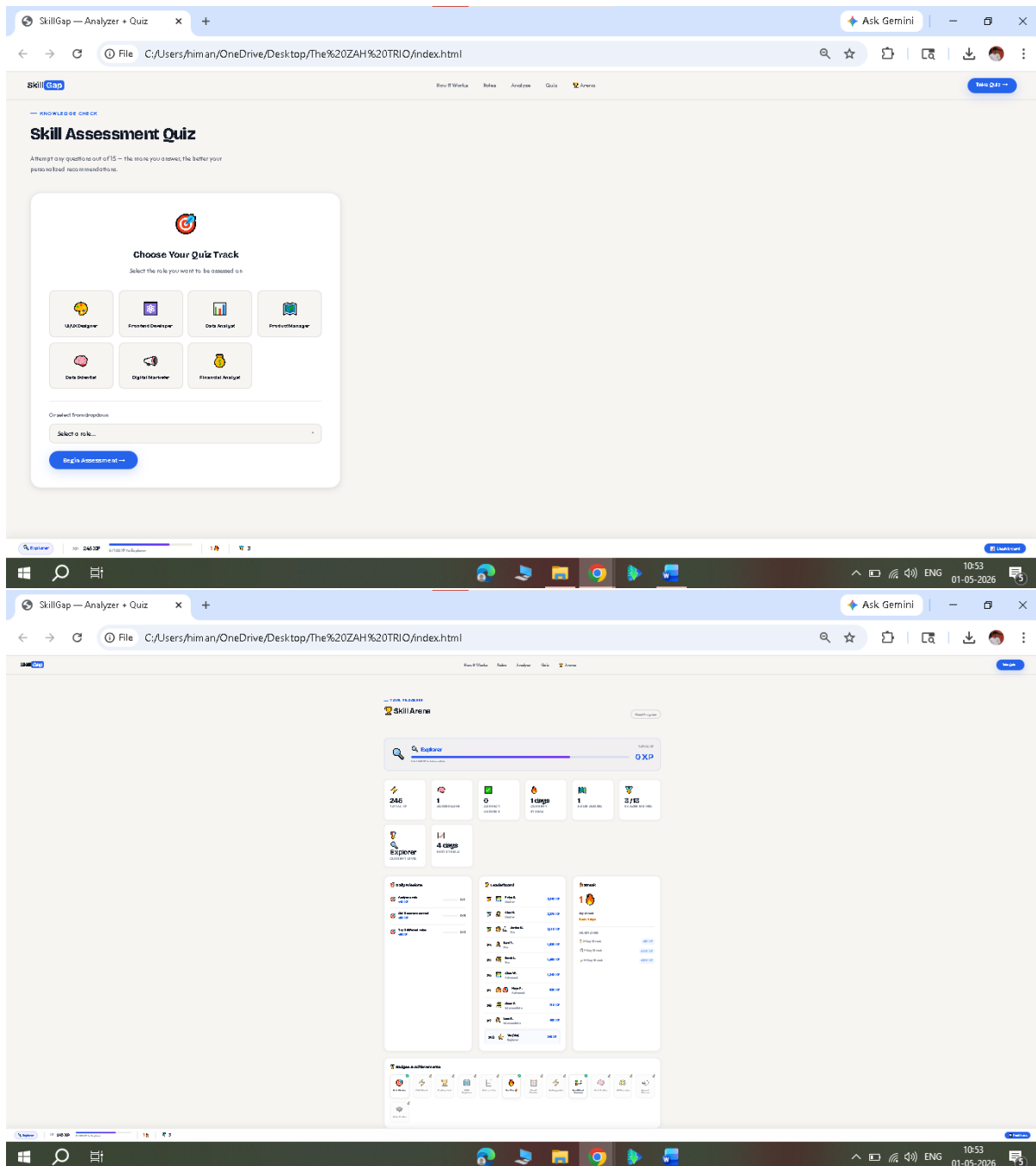
For developing the prototype, we used basic and effective web technologies.

- **HTML, CSS, and JavaScript** were used to design and build the frontend of the website, making it interactive and user-friendly.
- **Node.js and Express** were used for backend development to handle data processing and logic.

These tools helped in creating a functional and responsive system that demonstrates the working of the Skill Gap Analyzer.







3.5 Test and User Feedback User Feedback:

After developing the prototype, we tested it with a group of students and job seekers to understand their experience. Most users found the system helpful in identifying their skill gaps and appreciated the clear guidance provided. Users liked how easy it was to enter their skills or upload their resume and get instant results. They found the skill gap analysis simple to understand, which helped them realize what they were missing. The personalized learning roadmap was especially useful, as it gave them a clear direction on what to learn next instead of feeling confused. Many users also enjoyed the interactive features such as quizzes and leaderboard systems. These features made the learning process more engaging and motivated them to improve their skills regularly. The progress tracking feature also helped users see their improvement over time, which increased their confidence.

Observations:

During testing, we observed a few challenges faced by users. Some users felt that the system's skill comparison was not always very detailed, which sometimes affected the accuracy of results. A few users also found it difficult to decide which skills to input initially, especially beginners who were unsure about their current level. We also noticed that while the system was easy to use, adding more detailed explanations and suggestions could make it even more helpful for users.

Refinement Suggestions:

Based on user feedback and observations, we identified some improvements for the system. The accuracy of skill analysis can be enhanced by using better data and more advanced techniques such as AI-based evaluation. The system can also include suggestions for beginner users to help them identify their starting point. Additionally, more personalized recommendations and real-time updates based on industry trends can improve the effectiveness of the platform. Enhancing gamification features and adding more interactive elements can further increase user engagement and motivation.

Technology Stacks

Technology Stack:

The development of the Skill Gap Analyzer project involved the use of simple yet effective tools and technologies to create a functional and user-friendly system. The focus was on using technologies that are easy to implement while still delivering a smooth and interactive user experience.

Design Tools:

For designing the interface and overall layout of the system, tools like Canva and Figma were used. These tools helped in creating visually appealing designs, user interface layouts, and presentation elements. They made it easier to plan how the system would look and how users would interact with it.

Frontend Development:

The frontend of the project was developed using HTML, CSS, and JavaScript. HTML was used to structure the content of the web pages, while CSS was used to style and design the interface, making it visually attractive and responsive. JavaScript was used to add interactivity, such as handling user input, displaying results, and making the system more dynamic and engaging.

Backend Development:

For handling the logic and processing of data, Node.js and Express were used. These technologies helped in managing user input, analyzing skills, and generating results efficiently. The backend plays an important role in comparing user skills with predefined industry requirements and identifying the gaps.

Development Platform:

The entire project was developed using Visual Studio Code (VS Code), which served as the main development environment. It is a lightweight and efficient code editor that supports multiple programming languages and provides useful features like extensions, debugging tools, and an integrated terminal.

To improve development efficiency, extensions like Prettier were used for automatic code formatting, ensuring clean and consistent code. Additionally, tools like Live Server were used to run the project locally and view real-time changes during development.

IMPLEMENTATION

We used clean, meaningful code so screen readers and other tools could understand it.

We also tested it with keyboards and screen readers to make sure it works well for people with different needs.

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Skill Gap Analyzer</title>

<style>
/* ===== style.css + gamification.css ===== */

body {
  font-family: Arial, sans-serif;
  background: #f4f4f4;
  text-align: center;
}

header {
  background: #333;
  color: white;
  padding: 15px;
}

section {
  margin: 20px;
  padding: 20px;
  background: white;
  border-radius: 10px;
}

button {
  padding: 10px 15px;
  background: blue;
  color: white;
  border: none;
  cursor: pointer;
}

#gamification {
  background: #ffeaa7;
}
</style>

</head>

<body>

<header>
  <h1>Skill Gap Analyzer</h1>
</header>

<main>
<section>
```

```

    <h2>Enter Your Skills</h2>
    <input type="text" id="userSkill" placeholder="Enter your skill">
    <button onclick="addSkill()">Add Skill</button>
</section>

<section>
    <h2>Recommended Skills</h2>
    <ul id="recommendations"></ul>
</section>

<section id="gamification">
    <h2>Gamification</h2>
    <p>Points: <span id="points">0</span></p>
</section>

</main>

<script>
/* ===== app.js + gamification.js ===== */

let skills = [];
let points = 0;

function addSkill() {
    let input = document.getElementById("userSkill").value;
    if(input === "") return;

    skills.push(input);

    showRecommendations(input);
    updatePoints();

    document.getElementById("userSkill").value = "";
}

function showRecommendations(skill) {
    let list = document.getElementById("recommendations");

    let li = document.createElement("li");
    li.textContent = "Learn advanced " + skill;

    list.appendChild(li);
}

function updatePoints() {
    points += 10;
    document.getElementById("points").innerText = points;
}
</script>

</body>
</html>

```

RESULTS AND DISCUSSIONS

Key Outcomes:

Through the development of the Skill Gap Analyzer, a functional and user-friendly system was successfully created that helps users identify the gap between their current skills and industry requirements. The platform allows users to input their skills or upload their resume and provides clear results showing missing skills along with suggestions for improvement.

The system also includes features like personalized learning roadmaps, progress tracking, and gamification elements such as quizzes and leaderboards. These features make the learning process more structured, engaging, and goal-oriented. Overall, the platform provides users with a clear direction for skill development and helps them move closer to being job-ready.

Learnings:

During the development of this project, we understood that simply providing information is not enough; users need proper guidance and direction. This project highlighted the importance of user-centered design, where solutions are built based on real user needs and challenges.

We also learned that practical skill development is more important than theoretical knowledge in today's world. Working on this project improved our understanding of how to design simple yet effective systems that solve real-world problems. Collaboration and continuous feedback played a key role in improving the system and making it more useful.

Challenges Faced and How They Were Overcome:

One of the main challenges faced during the project was designing an accurate system to compare user skills with industry requirements. Since the system relies on predefined data, ensuring meaningful and relevant output was difficult. This was managed by simplifying the comparison logic and focusing on commonly required skills.

Another challenge was making the system engaging for users. Many platforms fail because they are not interactive enough. To overcome this, we included gamification features such as quizzes and leaderboards, which made the system more interesting and motivating.

We also faced difficulty in keeping the interface simple while still providing useful features. This was addressed by designing a clean and easy-to-use layout, ensuring that users can quickly understand the results and navigate through the system without confusion.

CONCLUSION

Role of Design Thinking in the Project

Design Thinking played an important role in shaping the development of the Skill Gap Analyzer. It helped us focus on real user problems and guided us in building a solution that is practical and user-friendly. Each stage of the process contributed to improving the overall system.

In the Empathize stage, we interacted with students and job seekers to understand their challenges. We found that many users are confused about which skills to learn and lack proper guidance for career development.

In the Define stage, we clearly identified the main problem — students are learning, but not the skills required by industries. This helped us focus on building a solution that bridges this gap.

In the Ideate stage, we explored different ideas and selected the concept of a Skill Gap Analyzer that not only identifies missing skills but also provides a clear roadmap for improvement.

In the Prototype and Test stages, we developed a working model and collected user feedback to improve the system. This helped us make the platform more simple, effective, and engaging.

Overall, Design Thinking ensured that the project remained focused on user needs and resulted in a solution that is not just functional, but also meaningful and helpful.

Future Scope of the Project

The Skill Gap Analyzer has strong potential for further development and improvement. In the future, advanced technologies like Artificial Intelligence can be used to improve the accuracy of skill analysis and provide more personalized recommendations.

The system can also be expanded to include job recommendations based on user skills and progress. Features like mentorship programs, discussion forums, and real-time industry updates can be added to make the platform more comprehensive.

With continuous improvements, this project can grow into a complete career guidance system that helps users move from learning to employment in a more structured and effective way..

REFERENCES

1. Books:

- Brown, T. (2009). Change by Design: How Design Thinking Transforms Organizations and Inspires Innovation. Harper Business.
- Norman, D. A. (2013). The Design of Everyday Things. Basic Books.

These books helped in understanding the concept of design thinking and user-centered problem solving, which played an important role in developing this project.

2. Websites & Articles:

- World Economic Forum. (2023). Future of Jobs Report.
- Coursera & Udemy – Online learning platforms for understanding current skill trends.
- LinkedIn Learning – Insights into industry-required skills and career development.

These sources provided useful information about skill demand, industry expectations, and current learning trends.

3. Research Papers:

- World Economic Forum Reports on skill gaps and employability trends.
- Various online research articles on skill development and career readiness.

These references helped in understanding the real-world impact of the skill gap and the importance of practical learning.

APPENDIX

• Interview Transcripts:

General Experience with Skill Development

Interviewer: How has your overall experience been with learning skills for your career?

Rahul: It's confusing most of the time. There are so many courses available online, but I don't know which skills are actually important for jobs. I end up learning random things without a clear goal.

Understanding Skill Requirements

Interviewer: Do you feel you have clarity about what skills industries require?

Priya: Not really. Colleges mostly focus on theory, and we don't get proper guidance about real-world skills. I often feel unsure about what I should learn next.

Features You Would Find Helpful

Interviewer: What kind of system or features would help you improve your skills?

Aman: A system that tells me what skills I am missing and gives a clear roadmap would be very helpful. Also, something that tracks progress and keeps me motivated would make learning easier.

• Survey Forms:

A survey was conducted using Google Forms with around 10 participants, including students and job seekers. The purpose of the survey was to understand the challenges faced in skill development and career preparation.

Some of the key questions included:

- "Do you know which skills are required for your desired job?"
- "What challenges do you face while learning new skills?"
- "Do you prefer structured learning paths or random courses?"
- "What features would you like in a skill development platform?"

The responses showed that most participants lack clarity about required skills and prefer a system that provides personalized guidance and a clear learning roadmap.